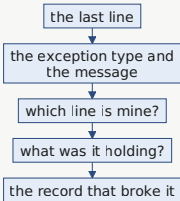


1 read a traceback



Read from the bottom up.

Line	Question it answers
Last	What went wrong, and on what value
Naming your file	Where you edit
Above that	How you got here

Three questions every time: exception type, my line, the value it held.

Crux. the value it was holding is the question people skip, and it is the one that names the record.

2 return against print

```
def f(r): print(r["order_id"])    # hands back None
def g(r): return r["order_id"]   # hands back the value
```

print is for the human at the screen. return is for the next line of code.

Crux. if the caller needs the answer, the function returns it.

3 catch narrowly

```
try:
    value = int(raw)
except ValueError as e:
    rejects.append({"id": rid, "reason": str(e)})
```

Catch	What it survives	What it hides
except ValueError	The failure you predicted	Nothing
except:	Everything	Every failure you did not predict

Crux. a crash costs an hour, a plausible wrong number costs a quarter.

4 the reconciliation

```
assert len(clean) + len(rejects) == len(rows)
```

Run it every time, especially when you expect it to pass.

Crux. input equals clean plus rejected, or something disappeared and you do not yet know what.

5 the four errors of Day 2

Error	It means
TypeError: 'NoneType' object is not subscriptable	A function printed instead of returning
ValueError: invalid literal for int() with base 10: 'x'	A word arrived where a number belonged
FileNotFoundError: [Errno 2] ...	The path does not exist from where the kernel is running
JSONDecodeError: ... line N column M	The parser ran out of valid input at that position

Crux. every one of these names the thing it choked on. Read it before you edit.

6 opening files

```
with open(path) as f:
    ...
```

The close is guaranteed, including when your code raises inside the block.

Paths are relative to the running kernel, never to the file browser.

Crux. FileNotFoundError prints the exact path it tried. Read it out loud and you find your own typo.

7 CSV and JSON

	CSV	JSON
Agrees about	Rows, commas, a header	Types and nesting
Types	Everything is text	Numbers stay numbers
Nesting	Cannot express it	Native
Partial read	Possible	Never, it parses whole or fails whole

```
csv.DictReader(f)      # keys come from the header row
csv.DictWriter(f, fieldnames=[...])
json.load(f) / json.dump(obj, f)
```

Crux. everything a CSV agrees to is text, so conversion is a decision you make on purpose.

8 one pass, two files

```
clean.csv    what converted
rejects.csv  what did not, and why
```

Then reopen both and count.

Crux. writing a file is not finishing. Reopening it and counting the rows is finishing.

THE WORDS ON THIS SHEET

- Function** One decision, named, with one home, callable again
- return against print** print shows a person, return hands a value to the next line
- Traceback** The stack of calls that led to a failure, read from the bottom
- Named exception** Catching the failure you expected and nothing else
- raise** Refusing on purpose, with your own wording and the offending value
- Rejects log** One row per record you could not use, each with its reason